



Edge and semantic collaboration framework with cross coordination attention for co-saliency detection

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ABSTRACT

Abstract—Co-salient object detection (CoSOD) aims to identify salient objects that appear concurrently in a set of related images. The existing CoSOD methods emphasize mining consensus clues, while tend to overlook the implicit edge and semantic information expressed at various levels of image features. To address this limitation, we propose a novel edge and semantic collaboration framework (ESCF). ESCF employs high-level features to align semantic information from highly activated salient regions, while utilizing low-level features to refine edge contours for improved clarity. To eliminate distracting elements within images, we introduce a cross coordination attention module (CCAM) that calibrates channel weights in each parallel branch using global information. By leveraging interactive learning across various feature layers, the model improves its capacity to capture position sensitive details, effectively minimizing noise in the process. Furthermore, we design a consensus learning module (CLM) that focuses on spatial and channel information across multi-scale features in parallel, allowing it to extract consensus clues from the generated multivariate attention maps, which in turn helps the network infer co-salient targets and ultimately guide it towards accurate predictions. We evaluate our method on three challenging CoSOD benchmark datasets using four widely recognized metrics, and experimental results demonstrate that our approach outperforms existing CoSOD methods.

1. Introduction

Salient object detection (SOD) is a fundamental task, which focuses on identifying visually prominent objects or regions within an image. With the rapid advancement of this field, specialized branches have been developed for various scenarios, such as RGB-D SOD [1], RGB-T SOD [2], ORSI-SOD [3], VDT-SOD [4] and VSOD [5]. Meanwhile, co-salient object detection (CoSOD) has also gained increasing attention, with a particular focus on detecting co-salient objects that share common attributes across a given set of related images. Unlike traditional SOD tasks, CoSOD poses greater challenges as it requires not only identifying co-salient objects across different images, but also suppressing interference from other objects within the same image. Additionally, CoSOD needs to consider both local saliency and global similarity, as these are key characteristics of co-salient target, such considerations necessitate an evaluation of both the accuracy of object appearances within individual images and the consistency of object categories across different images. Due to its potential utility in various fields, such as crop sensing [6], co-segmentation [7], weakly supervised learning [8], image retrieval [9], and video foreground detection [10], CoSOD is

gaining increasing popularity, Fig. 1 shows some excellent models in recent years.

Convolutional neural networks (CNNs) have achieved remarkable success in computer vision, contributing significantly to co-salient object detection. Early CoSOD methods relied on CNNs for feature extraction, effectively generating prediction maps by exploring common cues within image groups. However, their limited receptive field restricts the capture of long-range dependencies and contextual information in complex scenes. To address this, researchers have introduced attention mechanisms like Transformer [11], DETR [12], T2T-ViT [13], and Swin Transformer [14], these models enhance feature understanding and prediction accuracy. By integrating the local feature extraction capabilities of CNNs with the global attention mechanisms of transformers, this hybrid approach has significantly improved CoSOD performance.

Although prior research has achieved significant progress in CoSOD, existing methodologies have primarily focused on extracting global consensus patterns, resulting in insufficient consideration of skip connection mechanisms. Furthermore, the highly complex and heterogeneous backgrounds inherent in CoSOD tasks, combined with the

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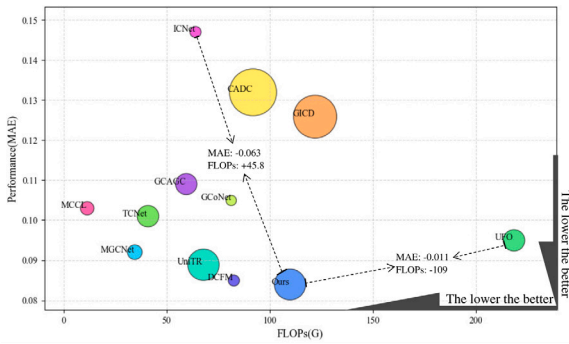


Fig. 1. We compared the performance (MAE) and FLOPs (G) of 11 existing representative CoSOD models and our model, where the larger the bubble, the larger the number of parameters of the model.

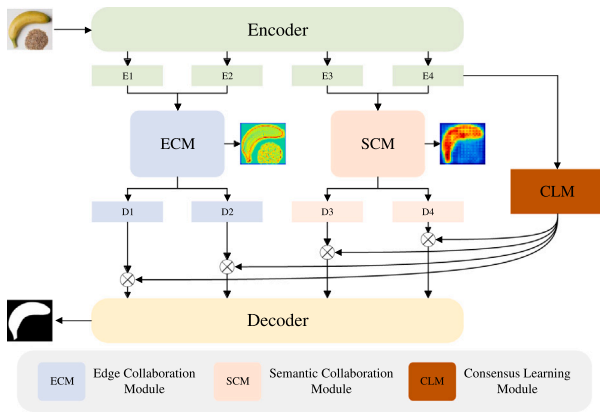


Fig. 2. Main idea of our proposed edge and semantic collaboration framework. The features extracted by the encoder are categorized into edge and semantic collaboration, which fusion with the co-salient features extracted from the highest-level features, the co-salient map is ultimately generated.

frequent presence of multiple distracting salient objects, often lead to erroneous identifications by conventional models. The strategic incorporation of attention mechanisms into skip connection operations effectively mitigates these issues through enhanced capability in modeling cross-level contextual relationships [15]. For instance, the UFO [16] model employs a transformer block with a depth of 4 for higher-level features, which captures global cues and strengthens the capability to model long-range pixel dependencies. However, focusing solely on global information is insufficient. Although some models can effectively identify salient objects, they often generate irregular saliency maps due to insufficient sensitivity to edge details. This limitation stems from the challenges of capturing high-level semantic relationships and texture information through manual features [17], which are vital for processing complex scenes. Moreover, in real-world scenarios, objects may appear in various orientations due to unpredictable factors. Therefore, enhancing the model sensitivity to directional changes is essential for improving its accuracy in detecting and segmenting objects.

Inspired by these observations, this paper introduces a novel cross coordination attention module (CCAM) that integrates horizontal branch, vertical branch and global branch into an enhanced cross-level learning framework by processing features from different levels in parallel. Our proposed strategy effectively employs high-level features to enhance global positional information and low-level features to refine local details, which leads to more effective feature integration. On these basis, we design a new edge and semantic collaboration framework (ESCF), as shown in Fig. 2, where the semantic collaboration module (SCM) enriches consensus semantics and saliency information, and the edge collaboration module (ECM) combines local edge information

with global position cues to extract salient edge features. To further refine consensus clues across images, we introduce a multi-scale attention mechanism within the consensus learning module (CLM), which addresses the limitations of single-scale consensus information.

Our main contributions are threefold as follows.

- We propose an edge and semantic collaboration framework (ESCF) for co-salient object detection. By synergizing cross-level semantic matching and edge refinement, ESCF locates targets via semantic correspondence while sharpening boundaries with edge cues, robustly distinguishing objects in cluttered scenes. This fusion significantly boosts accuracy in challenging scenarios.
- We introduce a cross coordination attention module (CCAM) that enhances fine-grained perception of spatial layout and object orientation. Importantly, this module integrates Cross-Level Learning to effectively combine multi-level and multi-scale features, thereby significantly improving the overall performance of the model.
- We present a novel consensus learning module (CLM) that utilizes multi-scale feature embeddings to enhance the extraction of co-category information, which also employing weighted attention to effectively identify salient regions within the local sub-branches.

2. Related work

2.1. Salient object detection

The salient object detection (SOD) task emulates the human visual system capability to identify the most prominent and representative objects in an image. This established computer vision task has seen significant advancements over the years. Early approaches primarily involved extracting features from localized regions of the image to generate target proposals [18] using superpixels [19] basic processing units for identifying salient objects. However, the sequential nature of regional feature extraction often resulted in lengthy processing times. Recent advancements have shifted focus towards fully convolutional network (FCN) [20], which have garnered considerable attention due to their impressive performance in image segmentation tasks. As discussed in PoolNet, the existing saliency detection models can generally be divided into five categories, which include single-stream [21], multi-stream [22], side-fusion [23], U-shaped [24], and multi-branch [25] architectures. Single-stream models generate saliency maps using simple convolutional networks, while multi-stream models extract multi-scale features by processing inputs at different resolutions. Side-fusion architectures fuse multi-level features at different stages to enhance prediction accuracy. U-shaped architectures, particularly those based on feature pyramid network [26] and U-Net [27], progressively aggregate finer details through a top-down path and are widely adopted in saliency detection tasks. Multi-branch architectures leverage multi-task learning to combine outputs from different tasks, improving salient object localization and edge detection. With advancements in the field, attention mechanisms and external information, such as edge and background cues, have been introduced, further enhancing model performance. These innovations allow saliency detection models to more accurately capture salient objects in complex scenes, significantly boosting their practicality and accuracy.

2.2. Co-salient object detection

Different from traditional SOD that detects salient objects within a single image, CoSOD focuses on identifying common salient objects in a group of images. In recent years, researchers have adopted a variety of supervised learning methods to explore the consistency within and between images, such as self-learning methods [28], joint attention between images and PCA projection [29], and co-clustering [30].

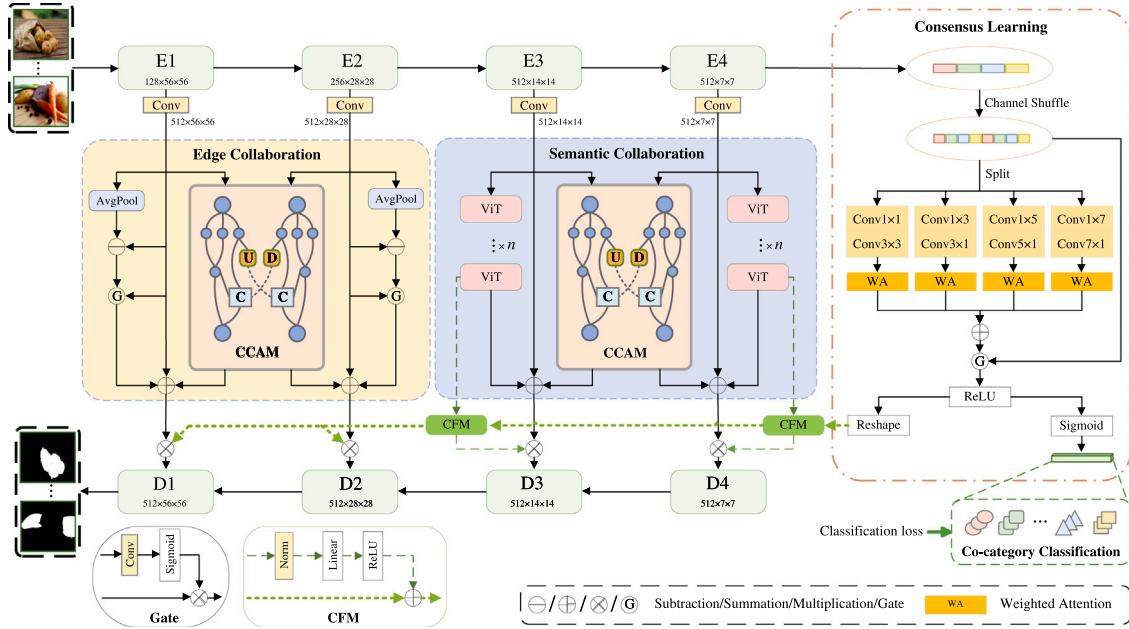


Fig. 3. The architecture of the proposed ESCF. First, we use a feature extractor to extract four-level basic feature embeddings. Then, we use the edge collaboration module and the semantic collaboration module to process the corresponding features respectively, and perform consensus learning on the highest-level features separately. The obtained consensus information is supplemented by CFM and multiplied by the features at each level. Based on the above, the final prediction graph is obtained.

There are also some models [31] that adopt a separated structure, that is, they first independently mine salient targets, and then use the learned consensus information to screen out common salient targets within the group. In addition, there are also models [32] that construct graph-based representations to simulate relationships between images, enabling more effective capture of co-salient targets with consistent features. At the same time, researchers are also exploring ways to mine hierarchical saliency information through features at different scales [33], different stages [34], or different levels [35]. Another research trend is to adopt semi-supervised [36] or unsupervised [37] methods to solve the CoSOD problem. As GCoNet [38] and GCoNet+ [39] proposed to explore the inter-group separability and intra-group compactness between two image groups, some research based on this has gradually emerged. For example, MCCL [40] uses the shared features within the group and the differences between groups to guide the model in generating feature map. In CoGEM [41], exchanging image groups containing different types of common targets to increase noise is also a way worth trying. These diverse methods are promoting the development of CoSOD technology and improving the accuracy and efficiency of identifying co-salient targets.

2.3. Coordinate attention

The attention mechanism is pivotal in deep learning, emulating human visual attention to enable models to focus on crucial information while disregarding less relevant parts. Notably, mechanisms like Squeeze-and-Excitation (SE) [42] have effectively enhanced model performance. However, traditional attention mechanisms often overlook position information, which is crucial for generating spatially selective attention maps. To address this limitation, Coordinate Attention (CA) [43] was introduced as a robust solution. CA integrates position information into channel attention, thereby preserving channel relationships while capturing long-range spatial dependencies and maintaining positional accuracy. This approach allows the model to more precisely locate and identify target regions. For instance, research [44] integrating CA into adversarial autoencoders has successfully guided networks to focus on salient targets in various directions, leading to higher-quality image results. In addition, research [45] enhances the ability to represent regions of interest by introducing CA in the

cross-layer information flow of the network. Nevertheless, in CoSOD scenarios, where numerous interference targets are present, relying solely on average pooling for information aggregation can introduce significant noise. To tackle this problem, we introduce CA to extract more precise position-sensitive information, which helps overcome the challenges posed by interfering targets in CoSOD images.

3. Proposed method

3.1. Overview

The proposed ESCF illustrated in Fig. 3, we employs a strategy for fusing features across different levels through upsampling and skip connections. After processing an image through the backbone network, four levels of feature maps are obtained and divided into two groups, high-level features are routed to the semantic collaboration module (SCM), while low-level features are directed to the edge collaboration module (ECM). Each module specializes in extracting distinct types of information, with the semantic collaboration module capturing rich semantic details and the edge collaboration module focusing on edge features, these features are then integrated through the decoder. The cross coordination attention module (CCAM) enhances the ability to perceive the location of salient targets by analyzing horizontal and vertical directions. It combines multi-level features to achieve accurate classification results and refine edge details. Additionally, the consensus learning model (CLM) is utilized to identify co-salient targets within image groups, generating co-category classification to ensure that features at each level are focused on the correct salient targets. Notably, the class mapping learned from high-level features in semantic collaboration further enriches the co-category information used by the CLM. Ultimately, these integrated features are fed into the decoder, which performs a linear combination to produce the final co-salient object prediction map.

Specifically, given a set of N images $P_i (i = 1, 2, \dots, n)$ containing co-salient objects, these images are fed into the encoder. The swin transformer backbone serves as the overall backbone for our network. The feature levels are denoted from low to high as $\{E_1, E_2, E_3, E_4\} \in \mathbb{R}^{N \times C \times H \times W}$, where N represents the batch size, C represents the number of channels, and H and W denote the height and width of the

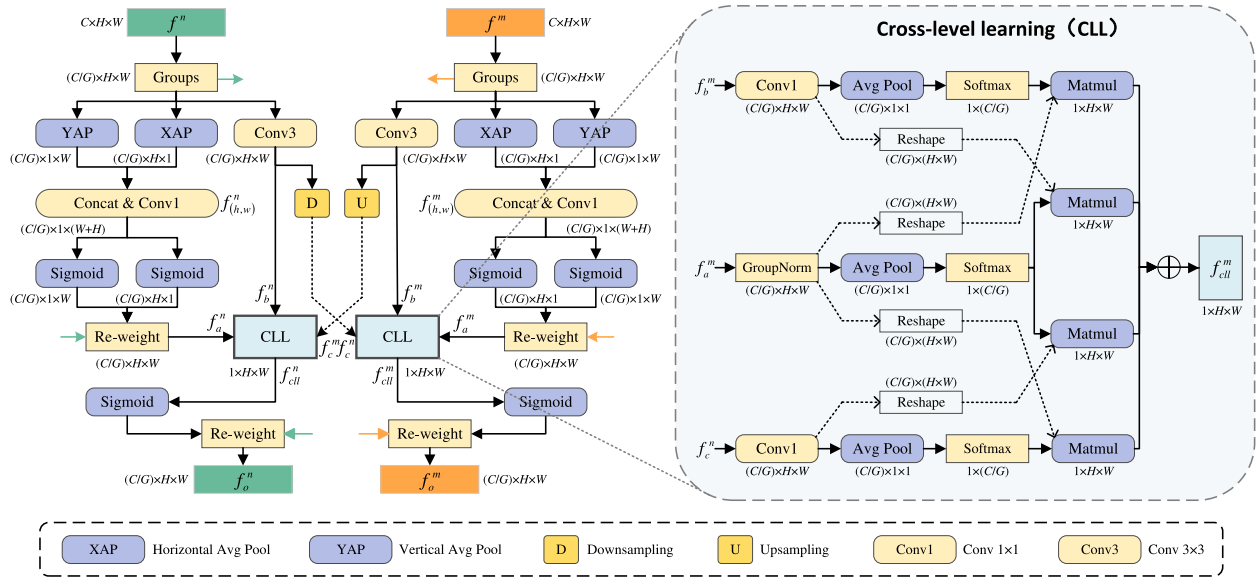


Fig. 4. The architecture of the proposed cross coordination attention module (CCAM). The integration of horizontal, vertical, and global processing across multiple branches, along with cross-level learning, which facilitates the exploration of global consensus representation across various branches and levels.

features, respectively. In this process, the features from the first two levels (E_1, E_2) are utilized in edge collaboration. This module extracts significant edge features by integrating local edge information with position-sensitive data, enhancing the capability to recognize and delineate edges more clearly. The latter two levels of features (E_3, E_4) are involved in semantic collaboration. This module combines position-sensitive information with feature correlations between image patches to capture more comprehensive semantic details. To fully leverage the semantic information contained in the top-level feature E_4 , it is processed by the consensus learning model to extract common category information. During this process, E_4 is shuffled across channels and divided into four sub-features, promoting interactions between features in different directions. Each sub-feature is then combined with multi-scale attention mechanisms to explore regional connections more deeply. These are aggregated to generate an attention map, which is subsequently refined through a gate mechanism to filter out noise. Finally, the refined attention maps are reshaped to match the dimensions of the features at each level, denoted as $\{g_1, g_2, g_3, g_4\} \in R^{N \times C \times H \times W}$, and fused with the corresponding features. This guides the extraction of co-salient targets, with co-category classification being supervised by classification loss.

3.2. Cross coordination attention module

In practical applications, objects of the same category frequently appear in diverse scenes and spatial contexts, exhibiting significant variations in orientation and position. For instance, a violin may be placed upright against a wall or laid flat on a table. These variations not only reflect environmental diversity but also pose two critical challenges, substantial orientation discrepancies (e.g., $\pm 90^\circ$ rotational differences) and stochastic positional variations under different imaging conditions. While CA [43] enhances spatial correlation modeling through coordinate encoding and ELA [14] improves computational efficiency via window-based processing, both approaches exhibit limitations in cross-level feature integration. Similarly, the EMA method [46] constructs cross-spatial aggregation paths to model long-range dependencies but primarily focuses on intra-level position encoding while overlooking inter-level semantic interactions. To address these shortcomings, we propose the cross coordination attention module, which enhances feature integration across different levels, facilitating more robust representation learning in complex spatial scenarios.

Based on the above considerations, we examine object position sensitivity in images from three perspectives, including horizontal branch, vertical branch and global branch. Meanwhile, we implement a parallel processing strategy in the cross coordination attention module, considering that the parallel substructure helps to avoid redundant sequential processing and reduce the network depth. The detailed structure of our CCAM is illustrated on the left side of Fig. 4. Given two features $\{f^n, f^m\} \in R^{C \times H \times W}$, we focus on one branch of the feature expansion process. We first reshape f^m to $f_g^m \in R^{(C/G) \times H \times W}$, where G is the number of groups. We then apply one-dimensional global average pooling along the horizontal branch and vertical branch. Specifically, we use pooling kernels with spatial dimensions $(h : 1)$ and $(1 : w)$ to capture global information along these axes. The pooled outputs are concatenated and passed through a convolution layer to integrate the horizontal and vertical information, as described by:

$$f_{(h,w)}^m = \text{conv} \left(\text{concat} \left(\frac{1}{h} \sum_{i=0}^h f_g^m(h, i), \frac{1}{w} \sum_{j=0}^w f_g^m(j, w) \right) \right) \quad (1)$$

where h and w are the pixel values in the horizontal and vertical directions. $f_{(h,w)}^m \in R^{(C/G) \times 1 \times (W+H)}$ represents the feature maps on the horizontal and vertical branches. This combined feature map is then split into two separate feature, $f_h^m \in R^{(C/G) \times H \times 1}$ and $f_w^m \in R^{(C/G) \times 1 \times W}$. These features are processed using a sigmoid activation function and then multiplied element-wise with the original feature f_g^m . The resulting position coordinate information f_a^m can be expressed as:

$$f_a^m = f_g^m \times \text{sigmoid}(f_h^m) \times \text{sigmoid}(f_w^m) \quad (2)$$

where i and j represent the x and y coordinates on the axis, respectively. In the final branch of the model, we use a 3×3 convolution operation to capture global information, denoted as f_b^m . Additionally, we downsample the global information from feature f^n , resulting in feature f_c^n , which has the same dimensions as f_b^m . The multi-level features extracted from three parallel branches undergo synergistic integration through our Cross-Level Learning (CLL) module, and are re-weighted with f_g^m feature. The process is as follows

$$f_o^m = f_g^m \times \text{sigmoid}(CLL(f_a^m, f_b^m, f_c^n)) \quad (3)$$

In the CLL, we introduce three tensors corresponding to previously learned feature information. In the upper and lower branches of the model, features are first processed by a 1×1 convolution kernel

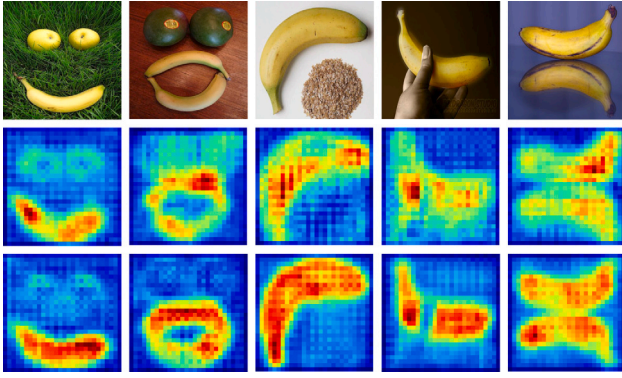


Fig. 5. Heat map analysis of edge and semantic collaboration. The first row shows the selected several representative original input images from the banana image group. The second row shows the responded maps without ECM and SCM. The third row shows the responded maps of the entire model.

to achieve channel-wise information integration. Next, we apply a two-dimensional global average pooling operation, reducing the image dimensions to (1,1) and encoding the global spatial information into a single descriptor. Unlike traditional pooling, which focuses on local areas, global average pooling captures relationships across the entire image, enhancing the ability to understand global context. These global descriptors are then activated using a softmax function to obtain a weight distribution for each channel. Unlike the upper and lower branches, the middle branch undergoes group normalization. After computing the weight of each channel, the upper and lower branches perform cross-level multiplication with the middle branch, followed by the addition and fusion of the four attention weights. The attention map obtained through cross-level learning and the jump connection re-weighted to obtain the final output f_o^m . The CCAM simultaneously mines the channel information of the image through three parallel branches, it then conducts cross-level learning to deeply explore the salient information from different regions, ultimately leading to a significant improvement in overall performance.

3.3. Edge and semantic collaboration framework

As the depth of the network increases, the ability of the model to perceive and understand images continues to improve. At shallower layers, the model primarily captures edge and local features, which provide rich texture and detailed information. As the network deepens, the model begins to abstract these details and shift focus toward global semantic features. These high-level features are rich in semantic content and are crucial for understanding the type, category, and location of objects in an image. In some datasets, the presence of complex scenes with objects that have indistinct edges and blend into the background poses challenges for segmentation. To address this, we propose an edge-semantic collaboration framework that integrates both edge features and semantic information, enabling more effective differentiation of co-salient objects from the background. Fig. 5 presents heat maps illustrating the intermediate processes of our two models. By comparing the second and third rows, it is evident that the ECM and SCM significantly broadens the activation range of salient objects. Notably, circular interference seen in the first and second columns is greatly reduced after applying the ECM and SCM, demonstrating that our module not only enhances feature representation but also improves the ability to accurately locate and interpret key regions within the image.

3.3.1. Edge collaboration module

In deep learning models, shallow networks are usually responsible for extracting low-level features in images, such as local details such as edges, corners, and textures. Specifically, edge information is a basic

element in visual perception. They can capture brightness changes and subtle edge information in images to effectively calibrate the boundaries and shapes of objects. To fully leverage these features, we apply edge collaboration to the two lower-level features $\{E_1, E_2\}$. In Fig. 3, we use a pooling subtraction operation in each branch to extract two levels of edge features $\{edge_1, edge_2\}$. Average pooling suppresses high-frequency edge information due to its smoothing effect. By subtracting the smoothed features from the original ones, the residual map enhances edge details while reducing background noise, leading to improved spatial precision and more effective edge detection compared to the pooled features. The process is as follows:

$$edge_n = E_n \ominus AP(E_n), \quad n = 1, 2 \quad (4)$$

where $\ominus$ denotes element-wise subtraction, AP represents average pooling, and $edge_n$ represents the edge information of E_n .

To address potential redundancy and noise issues that may arise from extracting only edge information, we introduce a simple yet effective gate mechanism. This mechanism aims to filter out unnecessary noise and retain purer, more useful edge information. First, we apply a convolution layer to process the extracted edge features. The convolution performs channel-level feature fusion, allowing for reweighting of feature channels without altering the spatial dimensions of the feature map. This helps to smooth the edge features and reduce noise introduced during edge extraction. After processing with the convolution, we use the sigmoid activation function to generate the corresponding feature map, converts the convolution layer output into a gate signal. The gate signal controls the extent to which the original edge features are passed through. Thus, only the edge information deemed important or reliable by the gate signal is retained, while potential noise is suppressed. We represent this gate mechanism as:

$$gate_n = \text{sigmoid}(\text{conv}_{1 \times 1}(edge_n)) \times E_n, \quad n = 1, 2 \quad (5)$$

where $\{gate_1, gate_2\}$ represents the features after noise removal, and $\times$ denotes element-wise multiplication. We strategically incorporate two different levels of features into the CCAM to activate key areas in the image, significantly enhancing the perception of important parts. This process enables us to accurately identify and emphasize critical features within the image. Furthermore, we effectively fuse the position-sensitive information processed by CCAM with the edge information. The final features obtained through edge collaboration can be expressed as:

$$F_n = gate_n \oplus E_n \oplus CCAM(E_n), \quad n = 1, 2 \quad (6)$$

This fusion strategy retains the rich details of edge information while enhancing the recognition of edge significance through position-sensitive information, thereby ensuring its validity and accuracy.

3.3.2. Semantic collaboration module

In image processing and analysis, high-level features can usually help models capture more complex patterns, relationships between objects, and the overall structure of the scene. These features contain deep semantic information such as object category information and context understanding. Therefore, they play a vital role in visual tasks, especially object recognition and scene parsing. As shown on the right side of Fig. 3, we select two higher-level features $\{E_3, E_4\}$ for semantic collaboration, capturing global clues through the transformer block in their respective branches. Additionally, we feed the two levels of features into the CCAM to obtain location-sensitive information. The final features generated by the SCM can be described as:

$$F_n = ViT^4(E_n) \oplus E_n \oplus CCAM(E_n), \quad n = 3, 4 \quad (7)$$

where ViT^4 represents a transformer block with a depth of 4. The captured global cues are additively fused with the position-sensitive information, which strengthens the ability to represent co-salient target regions.

To enhance the semantic understanding of model, we introduce a trainable category embedding vector at the beginning of the sequence, it enters the transformer block together with the features for training. This embedding vector adds additional information to the consensus clues in the consensus learning model by learning rich information about the image category. To this end, we design collaborative feature module(CFM) in Fig. 3 to fuse the embedding vector and co-category classification to make it smoother and minimize noise. CFM will be introduced in detail in Section 3.4.

3.4. Consensus learning module

Our focus is on developing effective methods to identify co-salient objects within a given set of input images. As mentioned in [47], high-level feature layers typically contain rich semantic information, which is crucial for detecting co-salient objects that share common characteristics. Additionally, multi-scale feature fusion can further enhance the feature space and improve the representational capacity. For instance, the multi-branch structure used in the Inception network, where each branch has a different local receptive field, allows the network to capture image features at various scales, thereby enriching feature representation. However, simply aggregating these multi-scale features through addition or concatenation may result in information redundancy and increased processing load.

To address these issues and fully leverage multi-scale features, we designed the consensus learning model. As shown in Fig. 3, we carefully process the highest-level feature E_4 to achieve multi-scale feature extraction and refinement. First, we shuffle the channels of E_4 and split it into four branches, represented as $\{e_4^1, e_4^2, e_4^3, e_4^4\}$. This step helps to break the dependencies between features and enhances the generalization ability of the model. One of the branches is directly subjected to 1×1 and 3×3 convolution operations. The other three branches are fed into independent parallel asymmetric convolutions for multi-scale feature extraction, with convolution kernel sizes of 3, 5, and 7, respectively. The entire process can be expressed as:

$$\begin{aligned} \hat{e}_4^1 &= \text{conv}1 \times 1 (\text{conv}3 \times 3 (e_4^1)) \\ \hat{e}_4^2 &= \text{conv}1 \times 3 (\text{conv}3 \times 1 (e_4^2)) \\ \hat{e}_4^3 &= \text{conv}1 \times 5 (\text{conv}5 \times 1 (e_4^3)) \\ \hat{e}_4^4 &= \text{conv}1 \times 7 (\text{conv}7 \times 1 (e_4^4)) \end{aligned} \quad (8)$$

This multi-branch structure facilitates capturing information at multiple scales simultaneously, with each branch focusing on feature extraction at different scales, thereby enriching the feature representation.

Next, we apply the weighted attention (WA) to the four multi-scale features to generate the corresponding attention maps. These attention maps enable comprehensive extraction of salient regions across multi-scale sub-features without losing important information from the original image. To fuse the attention maps of multi-scale features, we use a weighted fusion method by assigning a learnable parameter $\mu_n \in (0, 1)$ to each attention map. Its initial value of each parameter is set to 0.25, providing a balanced starting point for the model to be adjusted and optimized during training through back propagation and gradient descent algorithms. The fused attention feature map T_c can be calculated using the following formula:

$$T_c = \sum_{n=1}^4 \mu_n \cdot SA(CA(e_4^n)) \quad (9)$$

where n represents different feature branches, SA, CA represent spatial attention and channel attention respectively. During training, these learnable parameters are updated based on the gradient of the loss function to minimize prediction error and enhance model performance. As training progresses, each parameter gradually converges to an optimal value, highlighting the significance of the corresponding attention map in the final fused features. This approach produces a feature map T_c

is rich in salient target information. The left side of T_c is reshaped to match the shape of the features at each level and is multiplied pixel-by-pixel with features at each level to enhance co-saliency in the target areas. On the right side of T_c , co-category classification is generated using the sigmoid function.

It is worth mentioning that in order to make the category information more accurate, we designed CFM based on the class labels learned with features in the transformer block. It adds the common category information to the class labels after passing them through linear layers and ReLU. In this way, the model effectively enhances the semantic perception of common category information and adaptively focus on the location of common salient targets.

In summary, the consensus learning model enables our approach to effectively extract and utilize salient target information from images, while generating feature representations enriched with common category information.

3.5. Loss function

We use classification loss to guide the gradient propagation for updating the parameters of co-category classification. Classification loss measures the discrepancy between the predicted category and the true category, and updates the network weights through the back-propagation algorithm. The process is outlined as follows:

$$L_{cls} = L_{ce}(q, \hat{q}) \quad (10)$$

Here, L_{ce} represents the cross-entropy loss, q and $\hat{q}$ represent the ground-truth and the predicted, respectively. Additionally, we use weighted binary cross-entropy loss to achieve precise segmentation between pixels.

$$\begin{aligned} L_{wbce} &= -\frac{1}{hw} \sum_{n=1}^h \sum_{m=1}^w \log(Q(n, m)) \nu g(n, m) \\ &\quad - (1 - g(n, m)) \log(1 - Q(n, m)) (1 - \nu) \end{aligned} \quad (11)$$

where h represents the height of the image and w represent the width of the image, the ground-truth mask is represented as $g(n, m) \in \{0, 1\}$, the $Q(n, m)$ denotes the predicted probability of result. ν represents the ratio of positive pixels to the total number of pixels in the images. Additionally, We utilize IoU loss to assess the accuracy of segmentation, the process is as follows:

$$L_{iou} = -\ln \frac{I(q, \hat{q})}{U(q, \hat{q})} \quad (12)$$

where I represents the area of overlap between the predicted and the ground-truth, U represents the total area of the predicted and the ground-truth. The IoU loss is usually used to measure the similarity between the predicted and the ground-truth. By combining all relevant loss functions into a unified objective function and implementing training across the entire model architecture, we achieve comprehensive optimization of the system. This can be expressed as:

$$L_{total} = L_{cls} + L_{wbce} + L_{iou} \quad (13)$$

4. Experiments

4.1. Evaluation datasets and metrics

Dataset. We evaluate our model on three challenging datasets: CoSal2015 [48], CoSOD3k [49], and CoCA [50], which are widely used for benchmarking Co-SOD methods. The CoSal2015 dataset contains 2015 images spanning 50 different categories, designed to simulate real-world visual scenes where each group of images features objects captured from diverse scenes and viewpoints. CoSOD3k, the largest Co-SOD evaluation dataset to date, comprises 3316 images across 160 groups and 13 superclasses. Compared to CoSal2015, CoSOD3k presents more identical and different objects within each image, which

Table 1

Quantitative comparisons between the proposed method and 17 state-of-the-art methods. $\uparrow$ and $\downarrow$ mean that the higher and lower is better. The top three results are marked in red, blue, green, respectively. We use different symbols to mark different backbones, where $\dagger$ represents vgg, $\ddagger$ represents pvt, and $*$ represents swin transformer.

Model	Pub. & Year	ImageSize	CoSal2015				CoSOD3k				CoCA			
			MAE $\downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$	MAE $\downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$	MAE $\downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$
SSNM $\dagger$	AAAI 2020	224 $\times$ 224	0.102	0.794	0.843	0.788	0.120	0.675	0.756	0.726	0.116	0.482	0.741	0.628
GCAGC $\dagger$	CVPR 2020	224 $\times$ 224	0.095	0.832	0.888	0.823	0.094	0.779	0.864	0.789	0.109	0.517	0.757	0.666
GICD $\dagger$	CVPR 2020	224 $\times$ 224	0.071	0.844	0.887	0.844	0.079	0.770	0.848	0.797	0.126	0.513	0.715	0.658
ICNet $\dagger$	NeurIPS 2020	224 $\times$ 224	0.058	0.858	0.900	0.857	0.089	0.762	0.845	0.794	0.147	0.513	0.704	0.654
GCoNet $\dagger$	CVPR 2021	224 $\times$ 224	0.068	0.847	0.887	0.845	0.071	0.777	0.860	0.802	0.105	0.544	0.760	0.673
DeepACG $\dagger$	CVPR 2021	224 $\times$ 224	0.064	0.842	0.892	0.854	0.089	0.756	0.838	0.792	0.102	0.552	0.771	0.688
CoEGNet $\dagger$	TPAMI 2022	224 $\times$ 224	0.078	0.836	0.884	0.838	0.084	0.758	0.838	0.778	0.106	0.493	0.717	0.612
DCFM $\dagger$	CVPR 2022	224 $\times$ 224	0.067	0.856	0.892	0.838	0.067	0.805	0.874	0.810	0.085	0.598	0.783	0.710
TopicNet $\dagger$	KBS 2022	224 $\times$ 224	0.059	0.881	0.900	0.858	0.066	0.812	0.868	0.814	0.100	0.592	0.740	0.697
HrSSMN $\dagger$	TMM 2023	224 $\times$ 224	0.068	0.828	0.885	0.835	0.089	0.747	0.838	0.780	0.115	0.517	0.733	0.660
AGCNet $\dagger$	TMM 2023	224 $\times$ 224	0.055	0.860	0.907	0.868	0.066	0.801	0.880	0.830	0.116	0.546	0.745	0.686
TCNet $\ddagger$	TCSVT 2023	224 $\times$ 224	0.054	0.867	0.912	0.873	0.068	0.804	0.878	0.832	0.101	0.552	0.777	0.685
MCCL $\ddagger$	AAAI 2023	256 $\times$ 256	0.051	0.891	0.927	0.890	0.061	0.837	0.903	0.858	0.103	0.590	0.796	0.711
CoRP $\dagger$	TPAMI 2023	224 $\times$ 224	0.049	0.888	0.915	0.877	0.072	0.801	0.866	0.825	0.110	0.575	0.741	0.703
UFO $\dagger$	TMM 2024	224 $\times$ 224	0.064	0.865	0.906	0.860	0.073	0.797	0.874	0.819	0.095	0.571	0.782	0.697
CADC++ $\dagger$	TPAMI 2024	256 $\times$ 256	0.049	0.887	0.922	0.872	0.072	0.801	0.871	0.816	0.118	0.592	0.782	0.691
CONDA $\dagger$	ECCV 2024	224 $\times$ 224	0.049	0.894	0.926	0.890	0.067	0.807	0.873	0.832	0.102	0.600	0.774	0.717
UniTR $*$	TMM 2024	224 $\times$ 224	0.041	0.902	0.941	0.896	0.058	0.834	0.903	0.853	0.089	0.574	0.789	0.708
MGCNet $\dagger$	KBS 2025	224 $\times$ 224	0.048	0.887	0.923	0.877	0.057	0.827	0.897	0.837	0.092	0.600	0.786	0.708
Ours $*$	–	224 $\times$ 224	0.040	0.902	0.942	0.894	0.057	0.838	0.905	0.851	0.084	0.598	0.802	0.717

can interfere with detection results, making it more challenging. Although CoCA only contains 80 categories and a total of 1295 images, it has higher requirements for the dataset due to its complex background, occlusion, illumination changes, and the diversity and complexity of surrounding objects, which makes it the most challenging dataset among the three.

Metrics. We report quantitative results using four widely used evaluation metrics, including mean absolute error (MAE) [51], maximum F-measure (F_{β}^{max}) [52], maximum E-measure (E_{ξ}^{max}) [53], and structure-measure (S_a) [54].

4.2. Implementation details

The input image group consists of $N = 5$ images, with a batch size set to $8 \times N$. During training and testing, all images are resized to 224×224 pixels. We utilize the swin transformer as the backbone network, and all experiments are conducted on a single NVIDIA RTX 3090 GPU. Following the methods outlined in [55], we use the COCO-SEG dataset, which contains approximately 200,000 images grouped into 78 categories. Each image is provided with a manually labeled binary mask corresponding to a co-classified label. We employ the adam optimization algorithm to train the entire network, setting the exponential decay rates for the first-order and second-order moment estimates to 0.9 and 0.999, respectively. The initial learning rate is set to $2e-5$, and we reduce the learning rate every 20,000 epochs. The entire training process takes approximately 20 h, with a total of 60,000 epochs (see Fig. 6).

4.3. Comparison with state-of-the-art methods

To comprehensively evaluate our proposed framework, we compared it with 20 existing methods, including SSNM [56], GCAGC [57], GICD [50], ICNet [31], GCoNet [38], DeepACG [58], CoEGNet [29], DCFM [59], TopicNet [60], HrSSMN [61], AGCNet [62], UFO [16], TCNet [63], MCCL [40], CoRP [64], TCIC [65], UFO [16], CADC++ [66], CONDA [67], UniTR [68], MGCNet [69]. Given that co-salient object detection (CoSOD) methods have significantly outperformed single salient object detection (SOD) methods in recent years, we did not include single SOD methods in our comparison. To ensure a fair comparison, all co-saliency maps used in our evaluation were obtained directly from the original authors, and the performance metrics were directly referenced from the respective original papers.

Quantitative Comparison: As shown in Table 1, our model outperforms most of the comparison models across all three datasets. For instance, our model achieves superior performance in terms of the MAE and E_{ξ}^{max} scores across all datasets. Among the three datasets, CoCA is the most challenging because images often contain multiple distractors in addition to smaller co-salient objects. Our model achieves the best performance on all three metrics, significantly outperforming other methods with its stronger consensus ability. On two other datasets, except for the index which ranked second and third respectively, the other indexes are in the leading position. Additionally, the precision-recall (PR) curve and F-measure curve presented in the Fig. 6 further illustrate that our method consistently outperforms the other methods overall, especially on the CoSal2015 dataset.

Qualitative Comparison: Fig. 7 presents a visual comparison of our proposed method with other existing methods across three challenging image groups: banana, dolphin, and face powder. In the face powder image group, although most models can effectively identify co-salient targets, they are often influenced by interfering objects, leading to noise in the co-saliency map. Notably, in the first and fourth images of the dolphin group, ball objects are present. Most models mistakenly identify these objects as co-salient targets, whereas our method successfully avoids this misjudgment. Additionally, our approach effectively handles reflection interference. In the banana image group, our model accurately outlines the co-salient targets while remaining unaffected by distracting elements. Overall, compared to other methods, our approach not only effectively reduces noise but also excludes non-co-salient targets, providing more accurate detection of true co-salient targets.

4.4. Ablation study

We conducted ablation studies to evaluate the effectiveness of each component embedded in our models. The quantitative results for each module and their combinations are presented in Table 2. It is worth noting that our baseline approach employs the swin transformer as the backbone.

Effectiveness of CCAM: The cross coordination attention module is specifically designed to extract position-sensitive information from features at each level, enhancing the ability to perceive direction and position within both edge and semantic collaboration. The Table 3 illustrates the impact of integrating CCAM into our model and compares its performance with several existing attention mechanisms, including SE [42], ECA [70], CA [43], and ELA [71]. As shown in the table,

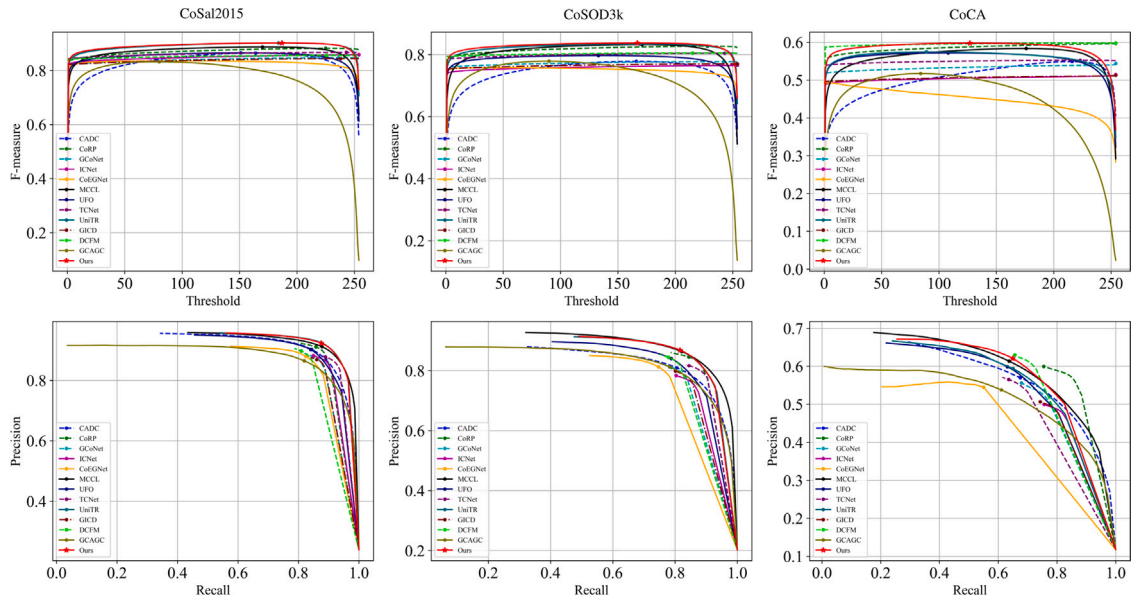


Fig. 6. Comparisons with other methods in terms of PR and F-measure curves on three benchmark datasets.

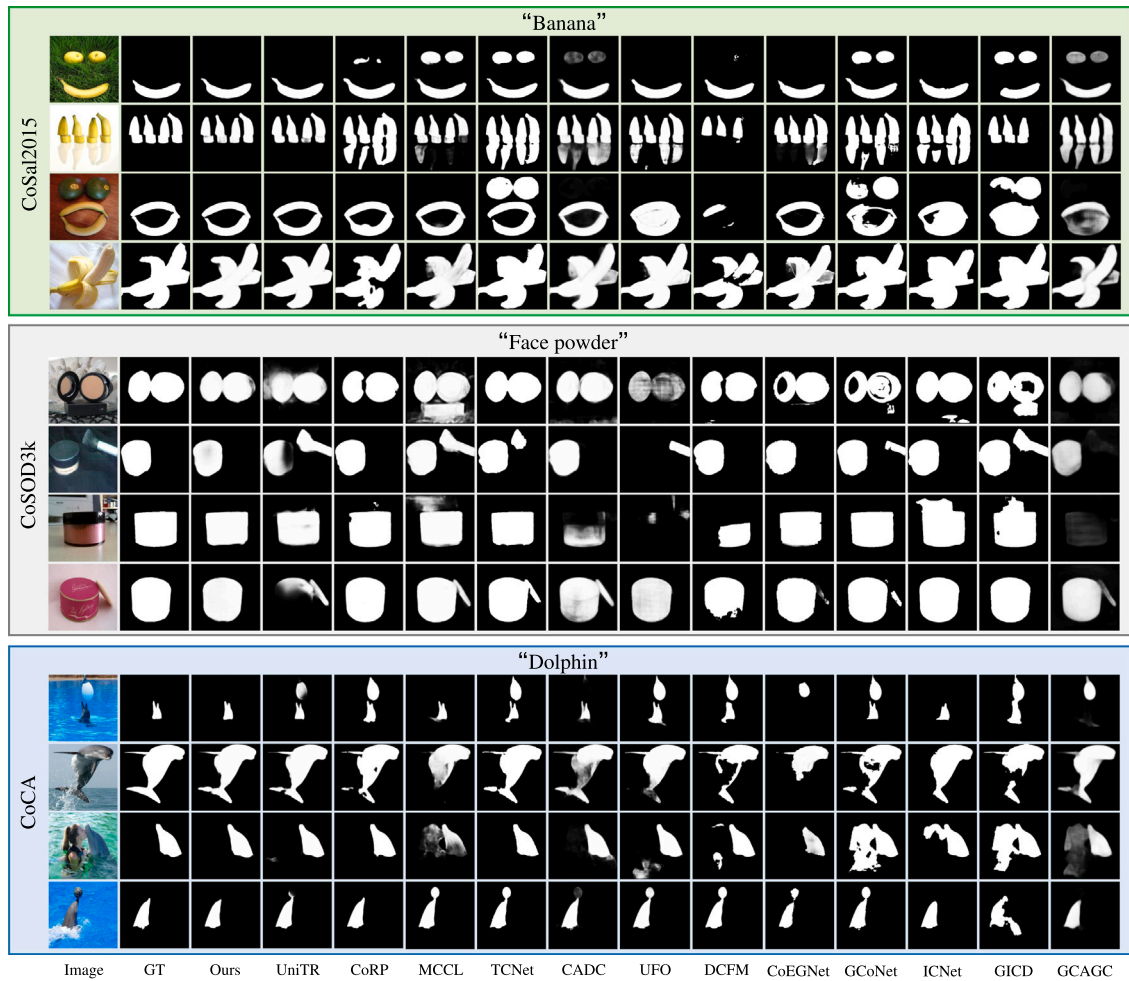


Fig. 7. Qualitative comparisons of our and other methods.

Table 2Ablation study of our proposed modules. The best results are in **bold**.

ID	Baseline	Modules			CoSal2015				CoSOD3k				CoCA			
		ECM	SCM	CLM	MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑	MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑	MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑
1	✓				0.074	0.840	0.901	0.849	0.084	0.772	0.868	0.809	0.117	0.463	0.731	0.639
2	✓	✓			0.063	0.859	0.913	0.861	0.068	0.801	0.881	0.826	0.099	0.523	0.757	0.670
3	✓		✓		0.049	0.881	0.928	0.880	0.063	0.819	0.893	0.838	0.095	0.561	0.770	0.697
4	✓			✓	0.056	0.868	0.920	0.871	0.069	0.799	0.883	0.826	0.100	0.504	0.752	0.662
5	✓	✓	✓		0.046	0.888	0.932	0.885	0.060	0.831	0.899	0.845	0.090	0.581	0.795	0.709
6	✓	✓		✓	0.054	0.873	0.924	0.875	0.063	0.820	0.893	0.839	0.094	0.561	0.772	0.696
7	✓		✓	✓	0.043	0.898	0.940	0.892	0.059	0.834	0.903	0.848	0.088	0.589	0.798	0.713
8	✓	✓	✓	✓	0.040	0.902	0.942	0.894	0.057	0.838	0.905	0.851	0.084	0.598	0.802	0.717

Table 3Ablation experiments of different attention mechanisms and CCAM. The best results are in **bold**.

ID	Strategy	CoSal2015				CoSOD3k				CoCA			
		MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑	MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑	MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑
1	SE	0.046	0.887	0.933	0.886	0.061	0.825	0.899	0.844	0.088	0.580	0.798	0.709
2	ECA	0.043	0.897	0.936	0.891	0.060	0.828	0.898	0.843	0.087	0.584	0.793	0.710
3	CA	0.042	0.898	0.939	0.892	0.059	0.833	0.902	0.848	0.088	0.584	0.789	0.710
4	ELA	0.044	0.892	0.936	0.890	0.060	0.828	0.899	0.847	0.089	0.586	0.791	0.715
5	CCAM	0.040	0.902	0.942	0.894	0.057	0.838	0.905	0.851	0.084	0.598	0.802	0.717

Table 4Ablation study for different parts in ECM. We conduct the Quantitative comparison between the model baseline, with (w/) CCAM, without (w/o) CCAM, with (w/) ECM. The best results are in **bold**.

ID	Schemes	CoSal2015				CoSOD3k				CoCA			
		MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑	MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑	MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑
1	Baseline	0.074	0.840	0.901	0.849	0.084	0.772	0.868	0.809	0.117	0.463	0.731	0.639
2	w/ CCAM	0.066	0.851	0.907	0.857	0.073	0.787	0.877	0.816	0.104	0.497	0.741	0.656
3	w/o CCAM	0.064	0.853	0.908	0.859	0.069	0.798	0.879	0.825	0.102	0.502	0.740	0.664
4	w/ ECM	0.063	0.859	0.913	0.861	0.068	0.801	0.881	0.826	0.099	0.523	0.757	0.670

Table 5Ablation study for different parts in SCM. We conduct the Quantitative comparison between the model Baseline, with (w/) CCAM, with (w/) TB, with (w/) SCM.GL means only using Transformer Block. The best results are in **bold**.

ID	Schemes	CoSal2015				CoSOD3k				CoCA			
		MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑	MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑	MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑
1	Baseline	0.074	0.840	0.901	0.849	0.084	0.772	0.868	0.809	0.117	0.463	0.731	0.639
2	w/ CCAM	0.064	0.853	0.911	0.852	0.073	0.790	0.878	0.813	0.104	0.506	0.743	0.657
3	w/ TB	0.054	0.876	0.926	0.874	0.066	0.814	0.891	0.832	0.096	0.561	0.764	0.692
4	w/ SCM	0.049	0.881	0.928	0.880	0.063	0.819	0.893	0.838	0.095	0.564	0.770	0.697

our CCAM consistently outperforms the other methods across all three datasets, with the most notable improvements observed on the CoCA dataset, an increase of 0.4% in MAE, 1.5% in F_{β}^{max} , 1.0% in E_{ξ}^{max} , and 0.6% in S_a . To further analyze the contribution of CCAM, we provide the attention maps it generates and visually compare them with existing attention mechanisms, as shown in Fig. 8. From the quantitative and qualitative results, traditional methods have two main limitations in co-salient object detection, weak ability to distinguish co-salient objects in complex backgrounds and inadequacy in maintaining fine-grained object structures. Specifically, although some methods are able to partially locate co-salient regions, their attention maps often have structural discontinuities at object boundaries, such as the obvious edge damage of the banana and drum in column 5. In contrast, our cross-layer feature integration mechanism is able to generate geometrically consistent attention responses that are precisely aligned with the object contour and exhibit smooth edge gradient transitions.

Effectiveness of the ECM: Table 4 shows the ablation experiments of ECM components. Compared with the baseline, ECM helps the model achieve excellent scores in all three datasets. On the CoCA dataset, the four indicators are improved by 1.8%, 6.0%, 2.6%, and 3.1% respectively. In addition, to further analyze the impact of ECM on boundary detection, we present a comparison of detection results on the helmet image set in Fig. 9. The experiment demonstrates that the introduction of ECM significantly enhances the ability to detect edge

Table 6Ablation experiments of CFM with different connection strategies. LB means using only the left branch of CFM, while RB means using only the right branch of CFM. The best results are in **bold**.

ID	Schemes	CoSal2015			
		MAE ↓	F_{β}^{max} ↑	E_{ξ}^{max} ↑	S_a ↑
1	w/ LB	0.045	0.890	0.933	0.888
2	w/ RB	0.044	0.893	0.935	0.889
3	w/ LB and RB	0.043	0.898	0.940	0.892

details. In the second row of images, the shadow interferes with the model without ECM, causing blurring at the lower helmet boundary and reducing the integrity of the target area. In contrast, the model with ECM achieves more precise boundary localization and effectively suppresses background noise, leading to smoother edge transitions. This pattern remains consistent across multiple samples, further validating ECM effectiveness in improving boundary recognition accuracy.

Effectiveness of the SCM: To evaluate the performance of the SCM within our model, we performed ablation studies on its components and explored different connection strategies for the CFM. As shown in Table 5, Using TB and CCAM together can enhance the perception of salient objects and generate more accurate prediction maps. The integration of SCM resulted in significant improvements across all three datasets,

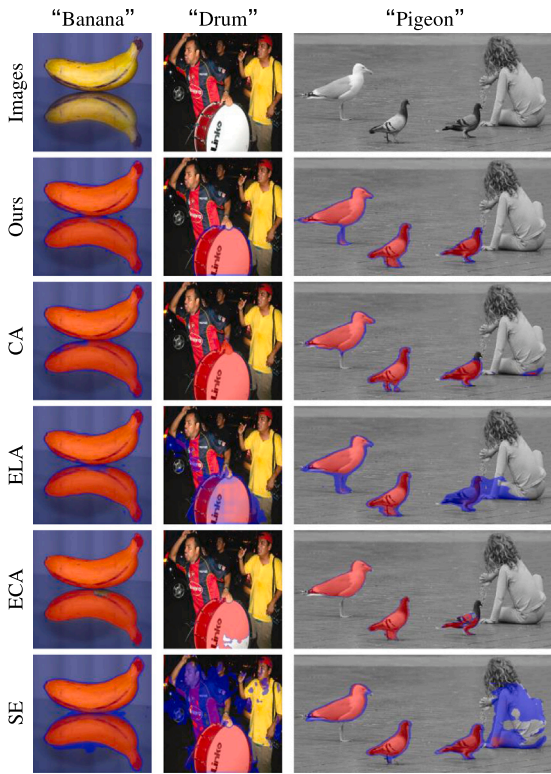


Fig. 8. Visualization of feature maps generated by our proposed CCAM and models adopting different attention methods. The blue part is the edge feature, and it appears red when focusing on a certain area.

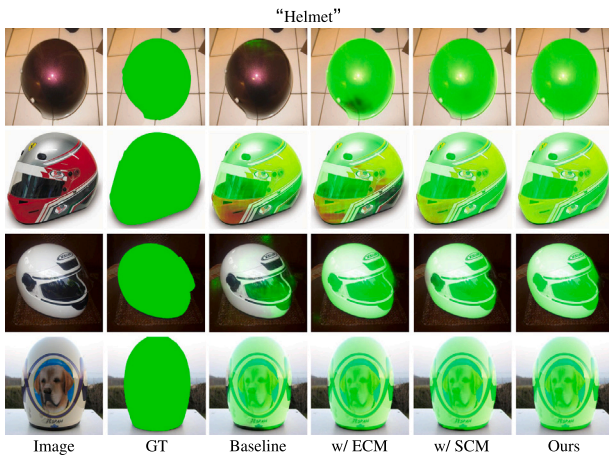


Fig. 9. Qualitative ablation studies of our ECM and ESM. Conduct the qualitative comparison between the baseline, with (w/) ECM, with (w/) SCM, and Ours entire model.

Table 7

Ablation experiments of CLM with different connection construction. Here a, b, c, d correspond to different connector architectures of CLM. The best results are in **bold**.

Strategy	Cosal2015			
	MAE ↓	$F_{\beta}^{max} \uparrow$	$F_{\zeta}^{max} \uparrow$	$S_a \uparrow$
a	0.046	0.888	0.932	0.885
b	0.044	0.891	0.933	0.887
c	0.044	0.892	0.934	0.888
d	0.040	0.902	0.942	0.894

with MAE scores increasing by 2.5%, 2.1%, and 2.2%, respectively. Table 6 further compares the connectors used in CFM, demonstrating

that class embedding information extracted at each layer enhances the ability to detect co-salient objects. Fig. 9 presents the detection results of our model applied to the helmet image group, both with SCM and without SCM. The results indicate that incorporating SCM significantly enhances the attention to the helmet area, leading to improved detection performance. This enhancement is primarily reflected in two aspects. First, SCM optimizes the synergy of semantic features, enabling the model to more accurately focus on the helmet region while reducing interference from irrelevant background elements. Second, SCM facilitates the fusion of high-level semantic information with low-level details, resulting in improved boundary clarity and structural integrity of the inferred helmet region. As demonstrated in the visualization results, the model with SCM more reliably captures key helmet features, minimizing information loss and reducing false detections. These findings validate the effectiveness of the SCM in improving the accuracy of detection.

Effectiveness of the CLM: Our consensus learning module combines multi-scale weighted attention, thoroughly considering the correlation between them to obtain more effective features. We conducted a systematic ablation study by gradually assessing the impact of adding various components, as illustrated in Fig. 10. Based on the quantitative detection results presented in Table 7, we can observe the impact of different connector architectures on model performance. Firstly, the proposed method partitions features into four distinct sub-features and processes them in parallel, which significantly enhances the robustness. Secondly, integrating a weighted attention mechanism across the four parallel sub-features significantly enhances the capacity to detect local salient regions within an image. This enhancement is attributed to WA to help the model capture local features more accurately by focusing on key areas of the image, thereby improving the overall performance. Specifically, the WA mechanism can dynamically adjust the weights between multiple sub-features, making full use of the information relevance at different levels to extract more consistent consensus clues. In addition, this mechanism can effectively reduce the information redundancy between different salient objects, allowing the model to more accurately focus on feature areas with global commonality during the reasoning process. In summary, these findings indicate that our consensus learning module, which processes multiple sub-sequences in parallel and effectively identifies local salient regions in a multi-scale space.

4.5. Failure cases

In Fig. 11, we present several failure cases of our method, along with those of the comparison method UniTR [68], in the context of baseball group detection. Upon analyzing these failure cases, we observed that both methods often misidentify salient targets in the images as batters. This misidentification primarily arises from the prominent positions of batters, uniform attire, and similar body movements, which collectively contribute to confusion in the identification process of algorithm. As a result, when there are numerous visually similar distracting objects in a set of images, our algorithm struggles to accurately detect the co-salient targets. We hypothesize that enhancing intra-group contrast may alleviate the effects of this issue. In co-salient object detection tasks, research efforts predominantly emphasize inter-group divergence analysis, while it is conventionally assumed that intra-group feature samples exhibit homogeneity. However, when certain samples contain co-salient targets that differ from the rest of the group, this assumption can lead to the problem mentioned above. Therefore, future research should concentrate on enhancing the attention to intra-group target differences, incorporating a more sophisticated intra-group contrast mechanism, and improving the accuracy of salient target recognition in complex scenes.

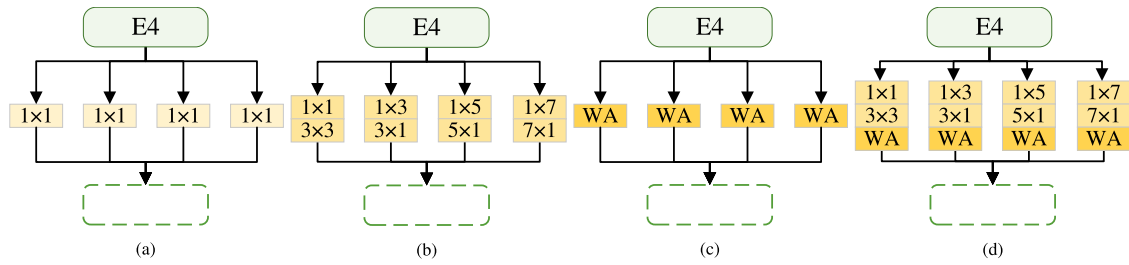


Fig. 10. CLM with different connector construction. (a) Splits the feature into four branches, each processed using 1×1 convolution. (b) Splits the feature into four branches, followed by multi-scale convolution. (c) Splits the feature into four branches, each processed using weighted attention. (d) Represents the final CLM configuration adopted in our model.

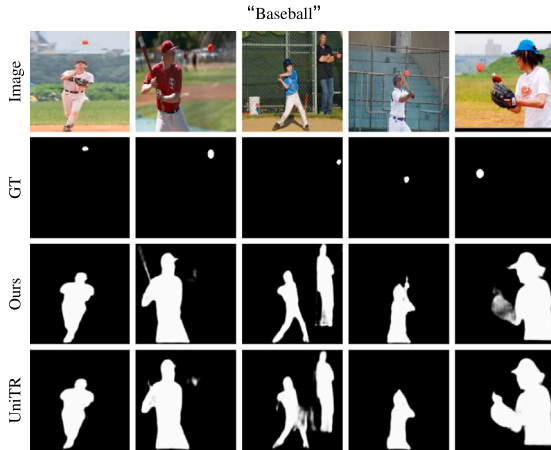


Fig. 11. Representative failure cases.

5. Conclusion

In this study, we aim to enhance the ability to recognize both edge and semantic information of salient targets in images by effectively integrating low-level and high-level feature extraction techniques. To achieve this, we propose a novel framework called edge and semantic collaboration framework (ESCF). Firstly, we introduce the cross coordination attention module (CCAM), which captures the positional information of each salient target in image features, and this module assists in extracting co-salient target features more accurately during the subsequent collaboration process. For low-level feature processing, we design an edge collaboration module (ECM) that employs a lightweight edge calibration mechanism to extract edge information from images, which is further refined by the positional guidance provided by CCAM to enhance the ability to detect target edges. For high-level features, we develop a semantic collaboration module (SCM) that leverages a transformer block to deeply explore the semantic information embedded in the image features. By aligning this semantic information with the position-sensitive data from CCAM, we significantly improve the understanding of salient targets semantics. Finally, we propose a consensus learning module (CLM) to enhance the extraction of co-category information. Our extensive experiments on three widely-used datasets demonstrate that the proposed ESCF method outperforms 17 state-of-the-art methods across four evaluation metrics. Furthermore, ablation studies validate the effectiveness of the proposed modules and highlight the superior performance of ESCF in the CoSOD task.

CRedit authorship contribution statement

Qiaohong Chen: Writing – review & editing, Writing – original draft, Resources, Conceptualization. **Qihang Wang:** Writing – review &

editing, Writing – original draft, Validation, Resources, Project administration, Formal analysis, Data curation. **Xian Fang:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. **Qi Sun:** Writing – review & editing. **Jinchao Zhu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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